RESEARCH REPORT

Physics-Guided Residual Learning for RANS Mean-Flow Correction

OVER THE NASA WALL-MOUNTED HUMP

A case-specific CFD-to-LES correction framework for separated turbulent flow

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NASA Wall-Mounted Hump Benchmark | SST-RANS + LES-Targeted Residual MLP

Abstract

Accurate prediction of turbulent flow separation remains difficult for conventional Reynolds-Averaged Navier-Stokes (RANS) turbulence models. This project develops a case-specific, physics-guided machine-learning workflow that corrects a baseline SST (k-ω) RANS prediction using higher-fidelity Large-Eddy Simulation (LES) reference data for NASA's wall-mounted hump benchmark. The hump produces a strong adverse pressure gradient and a downstream separated-flow region. A structured NASA grid was converted from PLOT3D format into OpenFOAM, and a steady incompressible SST-RANS simulation was performed using a turbulent boundary-layer inlet profile. The baseline CFD solution captured acceleration over the hump and lower-wall boundary-layer development but substantially underpredicted the separated reverse-flow region observed in NASA LES and experimental Particle Image Velocimetry (PIV) data.

A residual neural network was trained to predict the streamwise velocity correction, $\Delta U_x = U_{x,LES} - U_{x,RANS}$, using local RANS flow variables and spatial coordinates as inputs. To avoid an artificially easy pointwise validation, a continuous separated-flow region from $1.00 \le x/C \le 1.20$ was excluded from training. Within this hidden region, the full residual model reduced streamwise-velocity RMSE from 0.4253 for baseline RANS to 0.0140, corresponding to a 96.7% reduction. The AI-corrected field was also compared with independent NASA PIV measurements at hidden stations x/C = 1.03, 1.10, and 1.17. Relative to PIV, the correction reduced profile RMSE from 0.4451 to 0.1484, a 66.7% improvement. These results demonstrate that a residual machine-learning model can reconstruct LES-like mean-flow behavior within a held-out separated region of this benchmark. The method is not presented as a universal turbulence model or replacement for LES; it is a case-specific RANS-to-LES mean-flow correction framework.

Keywords: *turbulent flow separation; SST-RANS; large-eddy simulation; residual learning; NASA wall-mounted hump; physics-guided machine learning*

KEY RESULTS AT A GLANCE

0.4253 → 0.0140	0.4451 → 0.1484	$1.00 \le x/C \le 1.20$
LES hidden-region RMSE	**Independent PIV profile RMSE**	**Continuous withheld region**
96.7% reduction vs. baseline RANS	66.7% reduction vs. baseline RANS	A stronger spatial validation than random point splitting

1. Introduction

Turbulent flow separation is important in aerospace systems because it affects drag, lift, pressure recovery, control authority, inlet performance, and aerodynamic stability. Separation commonly occurs when a boundary layer encounters an adverse pressure gradient, causing low-momentum fluid near a surface to decelerate, reverse direction, and form a recirculating region.

RANS methods are widely used in engineering because they are substantially less computationally expensive than Large-Eddy Simulation (LES) or Direct Numerical Simulation. However, RANS models rely on turbulence closures that can struggle to represent separated shear layers and recirculation accurately. LES can provide higher-fidelity mean-flow predictions, but it requires substantially greater computational effort.

This project investigates whether machine learning can improve a low-cost RANS prediction by learning the difference between a baseline RANS solution and a higher-fidelity LES reference field. The selected test case is NASA's wall-mounted hump benchmark, which contains a curved hump along the lower wall of a channel. Flow accelerates over the hump, encounters an adverse pressure gradient downstream, and separates behind the hump. NASA's Turbulence Modeling Resource provides grid files and experimental PIV data for the no-flow-control configuration, while a corresponding LES mean-flow dataset is available for comparison [1–4].

The project objective was to develop and evaluate a physics-guided residual machine-learning model that corrects SST-RANS streamwise velocity predictions toward LES data for the NASA wall-mounted hump benchmark, then assess the corrected result against independent experimental PIV measurements.

SCOPE OF CLAIM This report evaluates a single benchmark geometry, Reynolds number, and inlet condition. The final model is a case-specific mean-flow correction framework, not a transferable turbulence closure or a replacement for LES.

Workflow overview. Physics-guided residual learning pipeline used in this study.

1	2	3	4
NASA benchmark data	**Baseline CFD**	**Residual learning**	**Independent validation**
Structured PLOT3D grid, LES mean field, and independent PIV measurements.	OpenFOAM SST-RANS solution with a turbulent boundary-layer inlet profile.	MLP learns $\Delta U_x = U_{x,LES} - U_{x,RANS}$ from local RANS fields and coordinates.	Continuous hidden region and PIV profiles evaluate generalization.

2. Benchmark Case and Reference Data

2.1 NASA Wall-Mounted Hump

The NASA wall-mounted hump is an essentially two-dimensional separated-flow benchmark based on a Glauert-Goldschmied-style hump mounted on the lower wall of a channel. Flow travels from left to right, accelerates over the forward portion of the hump, and separates on the downstream side.

The case was modeled using normalized reference quantities, $U_{in} = 1$ and $C = 1$, where U_{in} is inlet velocity and C is hump chord length.

$$\mathrm{Re\_C} = U_{in}\, C \,/\, \nu = 935{,}892$$

Therefore, the nondimensional kinematic viscosity used in OpenFOAM was:

$$\nu = 1 \,/\, 935{,}892 = 1.0685 \times 10^{-6}$$

Table 1. Reference scales and normalization used for the hump benchmark.

Parameter	Value	Role in this study
U_{in}	1 (nondimensional); 34.6 m/s (experimental)	Reference inlet velocity
C	1 (nondimensional); 0.420 m (physical)	Hump chord length
Re_C	935,892	Chord Reynolds number
ν	1.0685×10^{-6}	Nondimensional kinematic viscosity

2.2 NASA LES Data

The LES reference data contain mean velocity and Reynolds-stress quantities in the separated-flow region. The primary quantities used in this project were x/C, y/C, U_x/U_{in}, and U_y/U_{in}. The available LES field covered approximately $0.625 \le x/C \le 1.583$ with a tabulated resolution of 207 × 23. The LES field served as the high-fidelity target for residual-model training [2,3].

2.3 NASA Experimental PIV Data

Experimental PIV data were available in four overlapping regions downstream of the hump. The PIV files contain x/C, y/C, U, V, uu, vv, and uv. Streamwise PIV velocity values were normalized by the experimental inlet velocity, $U_x/U_{in} = U_x/34.6$. The PIV measurements were never used to train the AI model and were reserved for independent experimental validation.

3. CFD Methodology

3.1 Mesh Conversion

NASA supplied structured PLOT3D grid files at several refinement levels. The main CFD simulation used the hump2newtop_noplenumZ409x109.p3dfmt.gz grid. The NASA PLOT3D mesh was converted to OpenFOAM format using plot3dToFoam and then processed to create six physical boundaries. A mesh-view figure was not provided with the final analysis package, so boundary reconstruction and mesh statistics are documented in Tables 2 and 3.

Table 2. OpenFOAM boundary reconstruction for the converted NASA hump grid.

Boundary	Condition
inlet	Prescribed inlet velocity
outlet	Pressure outlet
symmetryFront	Spanwise symmetry plane
symmetryBack	Spanwise symmetry plane
topWall	Inviscid/slip wall
humpWall	No-slip lower wall and hump

Table 3. Final OpenFOAM mesh properties used for the baseline RANS simulation.

Mesh property	Value
Cells	44,064
Cell type	Hexahedral
Inlet faces	108
Outlet faces	108
Top-wall faces	408
Lower-wall/hump faces	408

The mesh contained no negative cell volumes and had acceptable skewness and non-orthogonality for preliminary RANS analysis. High-aspect-ratio cells were present near the wall, as expected for boundary-layer resolution.

3.2 RANS Solver and Turbulence Model

A steady incompressible SST (k-ω) RANS simulation was performed in OpenFOAM v13 using the incompressibleFluid solver framework. The SST model solves transport equations for turbulent kinetic energy (k) and specific dissipation rate (ω). It was selected because it is commonly used for wall-bounded aerodynamic flow and separated-flow prediction [5,6].

The governing mean-flow equations were:

$$\nabla \cdot \mathrm{U} = 0$$

$$\mathrm{U} \cdot \nabla \mathrm{U} = -\nabla \mathrm{p} + \nabla \cdot [(\nu + \nu_t)(\nabla \mathrm{U} + (\nabla \mathrm{U})^{\mathrm{T}})]$$

Here, ν_t is the turbulent eddy viscosity supplied by the SST closure.

3.3 Inlet Boundary Layer

A uniform inlet was initially used to verify solver operation, mesh conversion, boundary conditions, and ParaView visualization. This was replaced with an approximate turbulent boundary-layer inlet profile because the NASA experiment contains an established incoming boundary layer upstream of the hump.

A one-seventh-power-law profile was imposed:

$$\mathrm{U_x(z)} \;/\; \mathrm{U_{in}} = [\min(\mathrm{z} \;/\; \delta, 1)]^{1}/^{7}, \quad \delta = 0.07\mathrm{C}$$

The thickness $\delta = 0.07C$ was obtained from the LES reference documentation. Inlet turbulent quantities were initialized as $k = 1.5 \times 10^{-4}$ and $\omega = 0.32$. These values were suitable for a stable preliminary baseline but were not fully calibrated to experimental turbulence measurements.

3.4 Numerical Procedure

The simulation was stabilized using conservative numerical schemes and relaxation factors. The velocity field was initialized using potentialFoam, after which the steady RANS solver was advanced to a final pseudo-time/iteration level of 800. The final OpenFOAM result was exported to VTK format and processed in Python for interpolation, training-data assembly, and visualization.

4. Baseline RANS Performance

The baseline SST-RANS field captured several expected physical features: lower-wall boundary-layer development, acceleration over the hump, reduced near-wall velocity downstream of the hump, and a downstream low-momentum region. Acceleration over the hump was expected because the flow passage narrows locally. However, the baseline RANS solution did not adequately reproduce the separated reverse-flow region predicted by LES and observed in experimental PIV.

In the LES and PIV data, near-wall flow downstream of the hump becomes negative, $U_x < 0$. This negative streamwise velocity indicates local reverse flow inside the recirculation bubble. The baseline SST-RANS result remained mostly positive in the same region, indicating that it underpredicted separation strength and/or predicted reattachment too early.

Baseline Field Comparison

The field comparison below summarizes the central modeling gap that motivates the residual-learning correction: the baseline SST-RANS solution misses the LES-predicted near-wall reverse-flow region downstream of the hump.

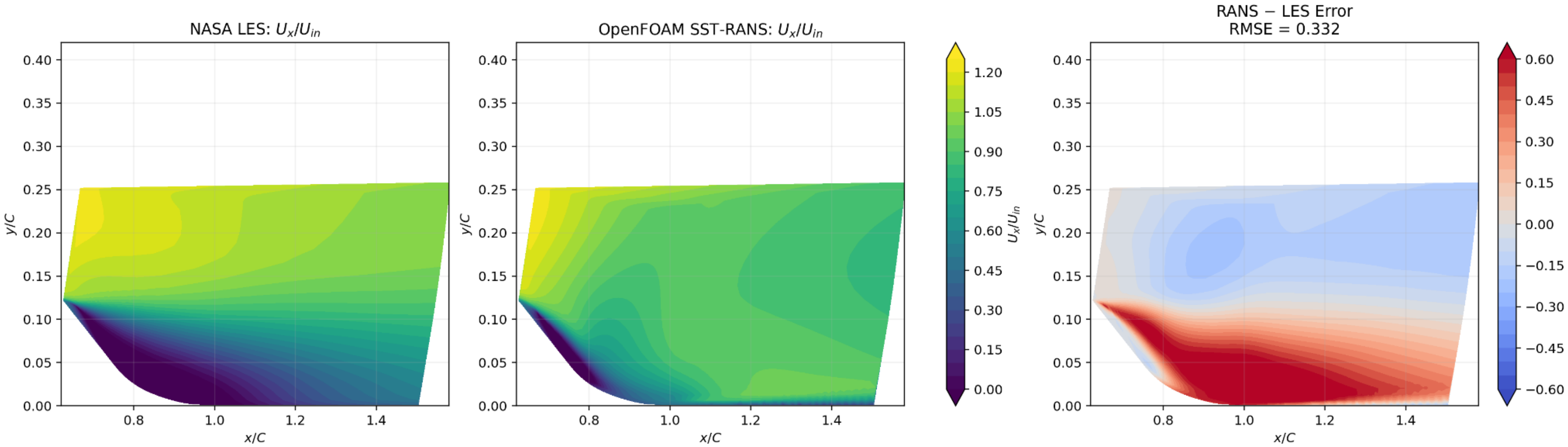

Figure 1. *NASA LES, OpenFOAM SST-RANS, and the RANS − LES streamwise-velocity error field. The baseline solution captures acceleration over the hump but underpredicts the downstream separated reverse-flow region.*

4.1 Baseline Velocity Profiles

Representative LES and baseline RANS profiles reinforce the contour-level observation. The greatest discrepancy occurs in the separated and recovery region, where LES contains negative or strongly reduced near-wall velocity while the RANS profile remains overly positive.

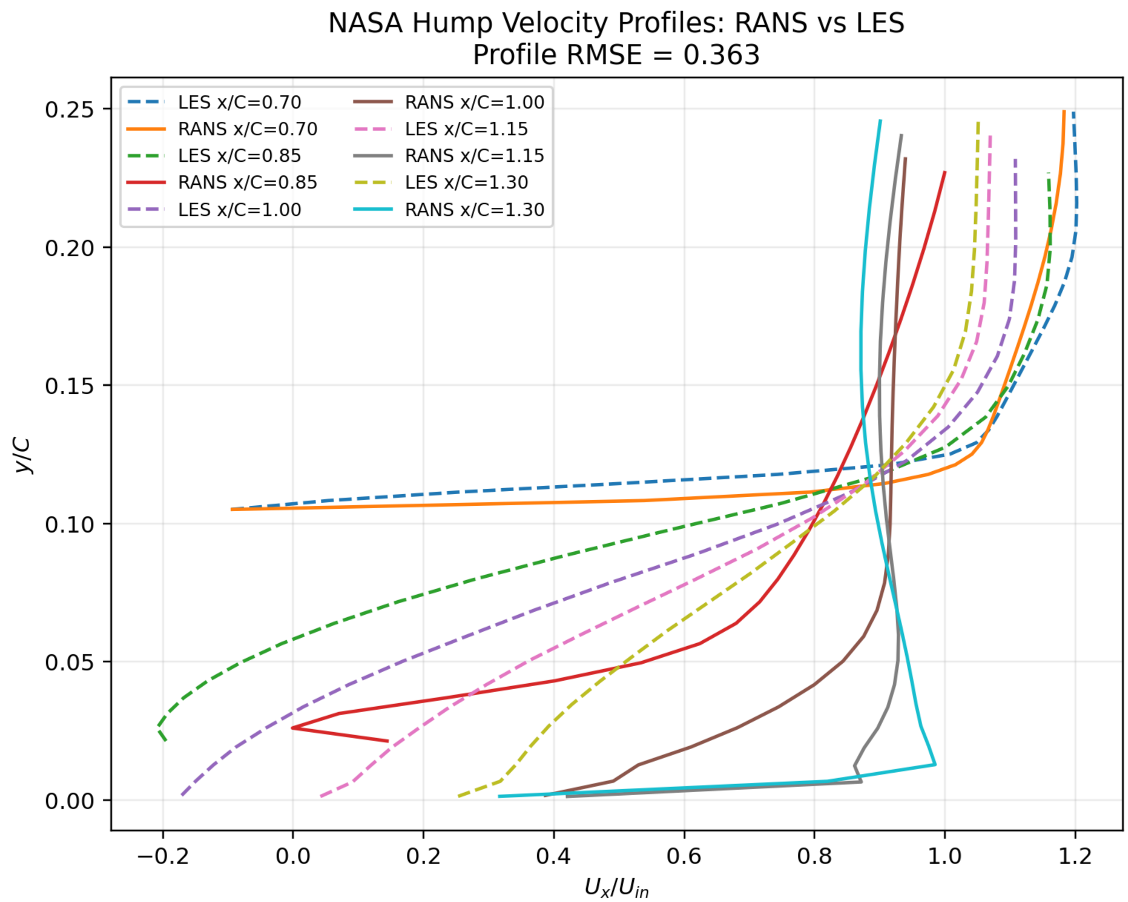

Figure 2. *NASA hump velocity profiles comparing baseline RANS with LES across several streamwise stations. The baseline solution does not adequately recover the LES reverse-flow behavior downstream of the hump.*

5. Residual AI Model

5.1 Learning Objective

Instead of asking a neural network to generate a flow field from scratch, the model learned a correction to an existing CFD solution. The target residual was defined as:

$$\Delta \mathrm{U_x} = \mathrm{U_x,LES} - \mathrm{U_x,RANS}$$

The corrected prediction was then calculated as:

$$\mathrm{U_x,AI} = \mathrm{U_x,RANS} + \Delta \mathrm{U_x,pred}$$

This residual-learning approach preserves the RANS prediction as a physics-based baseline while allowing the model to learn systematic error associated with turbulence closure and separated-flow prediction.

5.2 Input Features

The full residual model used eight local input features. The RANS fields were interpolated from the unstructured OpenFOAM mesh onto the LES sample locations. The wall-normal coordinate is written as z/C in the OpenFOAM feature set; figures use y/C for the same plotted wall-normal direction.

Table 4. Full residual-model input features.

Feature group	Inputs
Local mean flow	U_x,RANS; U_z,RANS; p_RANS
Turbulence closure state	k_RANS; ω_RANS; ν_t,RANS
Spatial coordinates	x/C; z/C

5.3 Neural-Network Architecture

The final blocked-validation model was a multilayer perceptron with the following structure:

$$8 \rightarrow 96 \rightarrow 96 \rightarrow 48 \rightarrow 1$$

The model used SiLU activation functions and predicted one scalar streamwise-velocity residual (ΔU_x). Training used normalized input features, normalized residual targets, mean-squared-error loss, AdamW optimization, and early stopping based on validation loss.

6. Validation Strategy

6.1 Why Random Point Splits Were Not Used

A random train-test split would place training and test points directly beside each other in the same smooth flow field. That arrangement can make a model appear more accurate than it truly is because it is largely interpolating between neighboring samples. Instead, a continuous downstream region was withheld from training:

$$1.00 \leq x/C \leq 1.20$$

This interval includes a major portion of the separated-flow and recovery behavior downstream of the hump. The model trained on LES data outside the interval and was evaluated only within the hidden region.

6.2 Ablation Models

Three AI models were compared to separate the contribution of geometry/location information from the contribution of local baseline CFD information.

Table 5. Ablation models used for blocked-region validation.

Model	Inputs	Purpose
Coordinates-only AI	x/C, z/C	Tests fixed-geometry/location learning
RANS-only AI	RANS flow variables without coordinates	Tests local CFD features alone
Full AI model	RANS flow variables plus x/C and z/C	Tests combined physical and spatial information

7. Results

7.1 Hidden-Region LES Validation

The full AI model was evaluated only within the completely withheld region, 1.00 ≤ x/C ≤ 1.20. The full model substantially outperformed both the coordinates-only and RANS-only models. Spatial location alone was useful for this fixed geometry, but the combination of local CFD variables and spatial information provided the strongest reconstruction.

Table 6. Hidden-region streamwise-velocity RMSE relative to NASA LES.

Model	Hidden-region U_x RMSE	Improvement vs. baseline RANS
Baseline SST-RANS	0.4253	—
Coordinates-only AI	0.0908	78.7%
RANS-only AI	0.1078	74.7%
Full RANS + coordinates AI	0.0140	96.7%

The AI-corrected field reproduced LES-like low-momentum and reverse-flow behavior in the hidden region that the baseline SST-RANS model missed. Figure 3 provides a profile-level overview of the corrected solution across the benchmark, and Figure 4 focuses specifically on the continuously blocked region.

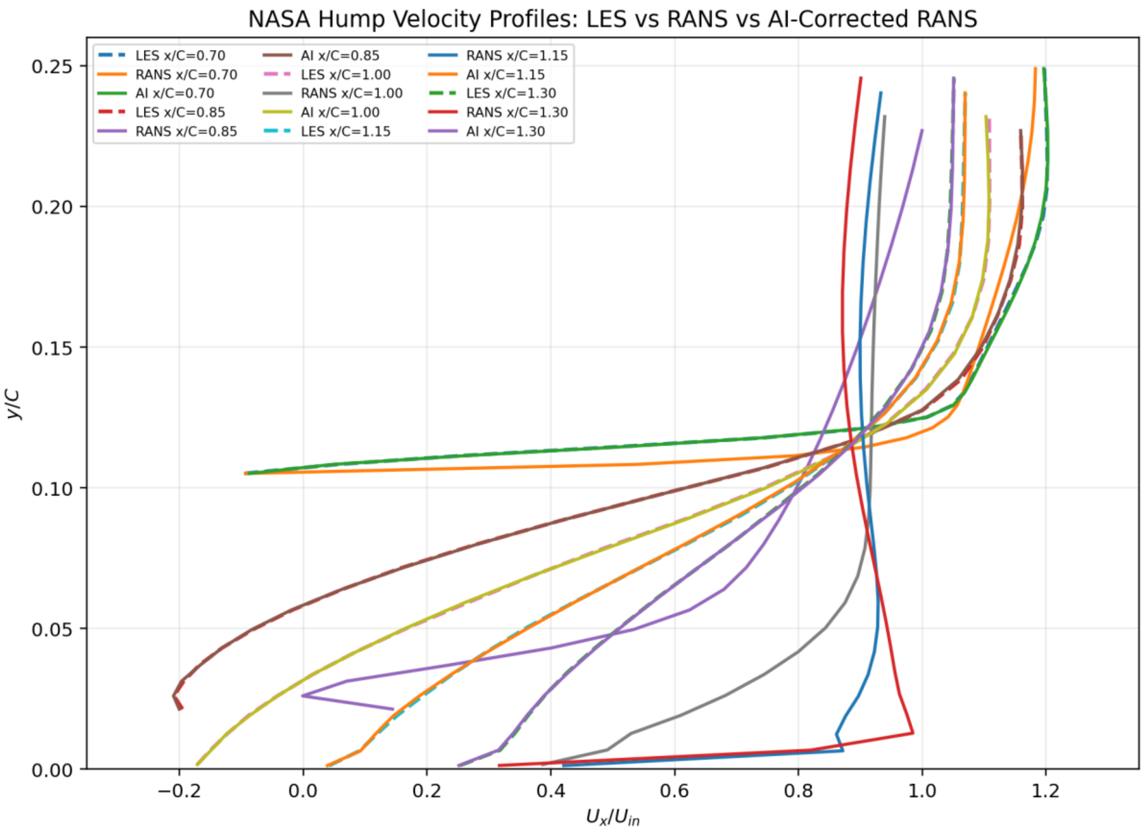

Figure 3. *NASA hump velocity profiles comparing LES, baseline RANS, and AI-corrected RANS. The corrected profiles recover the LES-like low-momentum behavior that is absent from the baseline solution.*

Blocked-Region Field Validation

The dashed lines indicate the continuous x/C interval removed from LES training. The error field remains small in the region hidden from the residual model during training.

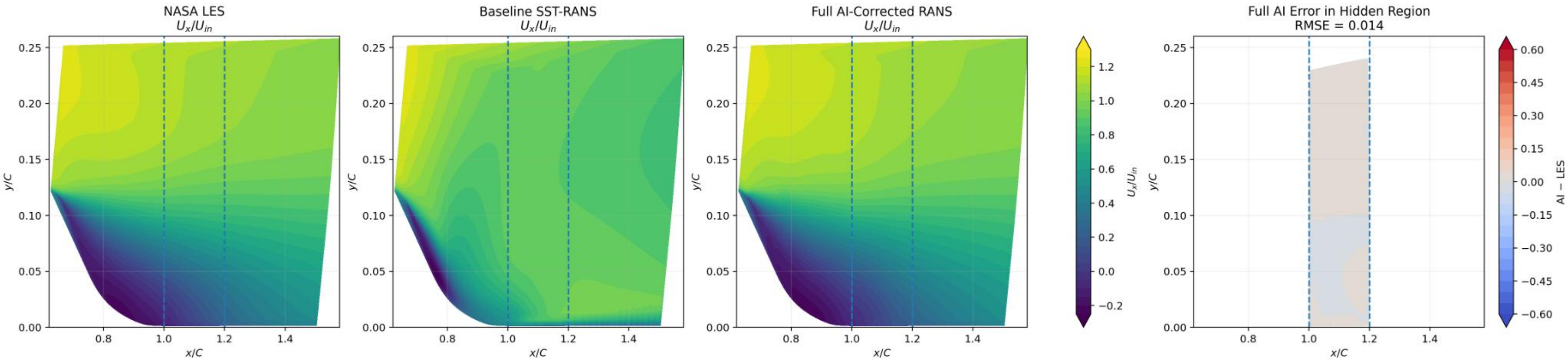

Figure 4. *Blocked-region validation of the full residual AI model. The interval $1.00 \le x/C \le 1.20$, marked by dashed lines, was excluded from LES training. Within this hidden region, the full AI model attains U_x RMSE = 0.014.*

7.2 Training Behavior

Training and validation loss both decrease by several orders of magnitude. The validation curve remains above training loss, as expected, but does not show an unstable divergence that would indicate catastrophic overfitting. Short loss spikes correspond to optimization updates and do not change the overall downward trend.

Figure 5. *Full residual-model training history for the blocked-region experiment. Normalized MSE loss decreases by several orders of magnitude for both training and validation data.*

7.3 Hidden-Station LES Profiles

Velocity profiles were extracted at x/C = 1.03, 1.10, and 1.17, all inside the hidden region. At these stations, baseline RANS remained too positive near the wall and did not capture the full separated-flow behavior. The AI-corrected profiles closely match the LES reference, including the low-momentum region and reverse-flow trend near the wall.

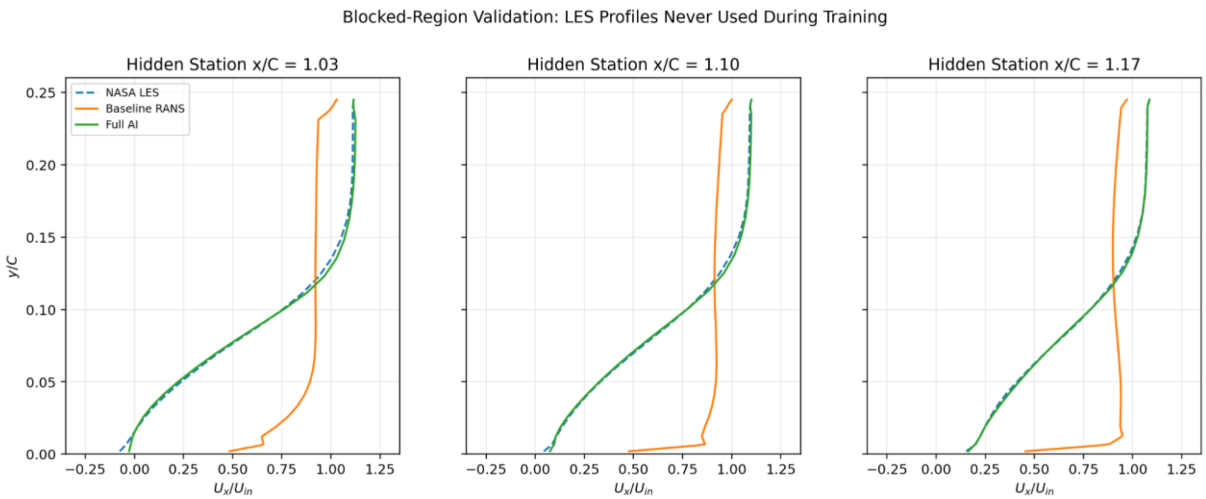

Figure 6. *Hidden-station LES profile comparison at x/C = 1.03, 1.10, and 1.17. These station profiles were never used during training; the full AI result closely matches NASA LES inside the blocked region.*

7.4 Independent Experimental PIV Validation

The final AI-corrected velocity field was compared with NASA PIV measurements at the same hidden stations: x/C = 1.03, 1.10, and 1.17. These PIV data were not used during AI training. The corrected profiles moved substantially closer to experimentally measured near-wall reverse-flow and recovery behavior.

Table 7. Independent NASA PIV validation inside the AI-held-out region.

Comparison	Streamwise-velocity RMSE	Improvement
Baseline RANS vs. NASA PIV	0.4451	—
AI-corrected RANS vs. NASA PIV	0.1484	66.7%

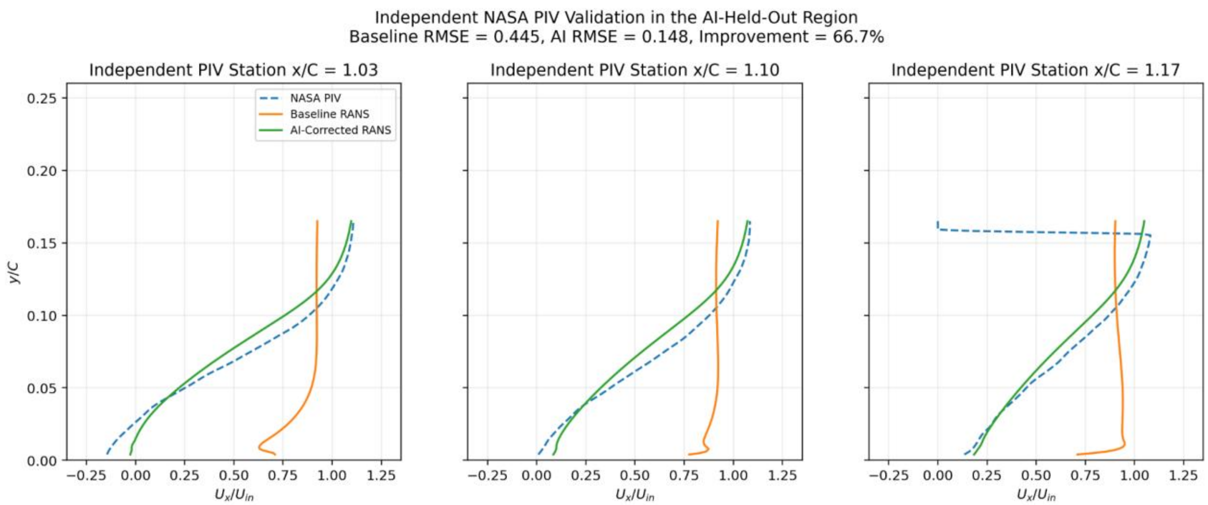

Figure 7. *Independent NASA PIV validation at hidden stations. Relative to baseline SST-RANS, the AI-corrected field reduces profile RMSE by 66.7% while better representing the measured separated-flow behavior.*

8. Discussion

The baseline RANS solution provided a stable, physically meaningful low-fidelity flow field, but it underpredicted the separated reverse-flow region behind the hump. This behavior is consistent with the known challenge of modeling separated turbulent flow using conventional eddy-viscosity RANS closures.

The residual machine-learning model corrected this systematic deficiency. The best result came from the full model, which combined local RANS variables with spatial coordinates. The coordinates-only model also performed well because the benchmark contains one fixed geometry and one smooth mean-flow field. Location is therefore naturally informative in this case. However, the full model's improvement over both coordinates-only and RANS-only variants indicates that local CFD information improved the reconstruction beyond purely geometric interpolation.

The PIV result is especially important because PIV was not used in training. While the model was trained using LES data from the same physical benchmark, it was evaluated against an independent experimental data source in a region excluded from LES training. The 66.7% reduction in PIV RMSE provides evidence that the AI-corrected field improved agreement with measured separated-flow behavior rather than merely reproducing the LES training data.

9. Limitations

- The model was trained and evaluated on one geometry, one Reynolds number, and one inlet condition; it should not be described as a universal turbulence model or a general replacement for LES.

- The model predicts mean streamwise-velocity correction only. It does not reconstruct instantaneous turbulence, time-dependent eddies, full Reynolds stresses, or a complete three-dimensional turbulent flow field.
- The RANS inlet turbulence quantities were initialized using practical but approximate values. A stronger CFD baseline could use a measured inlet profile, precursor boundary-layer simulation, improved turbulence initialization, and additional grid refinement.
- Spatial blocked validation is stronger than random point validation but remains interpolation within one physical case. Future work should test additional inlet conditions, grid levels, hump geometries, turbulence models, and Reynolds numbers.
- The baseline mesh had high-aspect-ratio near-wall cells. Although the mesh passed major quality checks, future work should include wall-distance and y^+ assessment together with a formal grid-convergence study.

10. Conclusion

This project developed a complete CFD-and-AI workflow for the NASA wall-mounted hump separated-flow benchmark. A NASA-provided structured grid was converted into OpenFOAM, boundary conditions were reconstructed, and a steady SST (k-ω) RANS simulation was performed using a turbulent boundary-layer inlet profile. The baseline RANS solution captured flow acceleration over the hump and downstream low-momentum behavior but underpredicted the experimentally observed separated reverse-flow region.

A residual neural network was trained to predict the LES correction to streamwise RANS velocity. In a continuous region withheld from training, the full model reduced LES-based RMSE from 0.4253 to 0.0140, corresponding to a 96.7% improvement relative to baseline RANS. When compared with independent NASA PIV data at hidden stations inside the withheld region, the AI-corrected field reduced streamwise-velocity RMSE from 0.4451 to 0.1484, a 66.7% improvement over baseline RANS.

The final result demonstrates that a case-specific, physics-guided residual model can substantially improve a baseline RANS mean-flow prediction in a separated-flow region. The project provides a foundation for future work involving multi-case training, divergence-aware loss functions, refined CFD baselines, and broader aerodynamic generalization studies.

Nomenclature

Key symbols and abbreviations used throughout the report.

Symbol / Term	Definition
C	Hump chord length
U_x	Streamwise mean velocity
U_{in}	Reference inlet velocity
Re_C	Chord Reynolds number
v	Kinematic viscosity
v_t	Turbulent eddy viscosity
ΔU_x	Residual streamwise-velocity correction
LES	Large-Eddy Simulation
PIV	Particle Image Velocimetry
RANS	Reynolds-Averaged Navier-Stokes

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